

The Continuous m -Center Problem on a Network

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In the continuous m -center problem on a network, m -centers are to be located along the arcs of the network so as to minimize the maximum (shortest) distance from any point on the network to a nearest center. The typical application scenario is to the location of emergency facilities in a transportation network. In this article, we propose a solution procedure for this hitherto unsolved problem. The procedure is a relaxation algorithm which may be viewed as a column and row generating procedure for the infinite size covering problems needed to solve the problem. The optimum-seeking procedure is expected to be practical for large-scale networks.

1. INTRODUCTION

Let $G = (V, A)$ be an undirected graph with a finite vertex set $V = \{v_1, v_2, \dots, v_n\}$ and a finite set A of arcs. To every arc $(i, j) \in A$, connecting v_i to v_j , associate a finite positive real number $b(i, j)$ representing its length. An arc may be considered as an infinite set of points. A point in arc (i, j) is specified by its distance from either of the endpoints v_i, v_j . Let G denote also the set of all points in the graph including both the set of vertices and all points in all arcs. Define $d(x, y)$ as the length of a shortest path between $x, y \in G$. Let $X_m = \{x_1, x_2, \dots, x_m\}$ denote a set of m points in G and let the quantity $d(X_m, y)$ be the generalization of $d(x, y)$ defined by

$$d(X_m, y) = \min_{x \in X_m} d(x, y). \quad (1)$$

Finally, let $l(X_m)$ be defined by

$$l(X_m) = \max_{y \in V} d(X_m, y). \quad (2)$$

We are now ready to formulate the “absolute m -center problem.”

Definition 1. For $m = 1, 2, \dots$, a set of points $X_m^* \subseteq G$ is a set of *absolute m -centers* of G , if for every $X_m \subseteq G$,

$$l(X_m^*) \leq l(X_m),$$

where $l(X_m)$ is given by (2). Furthermore, the quantity $r_m = l(X_m^*)$ is known as the *absolute m -radius* of G .

A common application of the absolute m -center problem is to the location of emergency service facilities on a transportation network, where x represents the location of such a facility and y represents the location of a demand point. For example, we may wish to locate m fire stations in a rural community in a manner that minimizes the maximum response time from the closest station to any farmhouse.

An important variant of the absolute m -center problem arises when demand for service occurs anywhere on the network (i.e., along the links as well as on the nodes). An obvious scenario for this problem is the previous fire-station problem in an urban context. Another example is an ambulance location problem where demand can occur anywhere on a street network. Mathematically, this "continuous m -center problem" is defined as in Definition 1, with $l(\cdot)$ replaced by $l'(\cdot)$, where

$$l'(X_m) = \max_{y \in G} d(X_m, y). \quad (3)$$

Our objective is to present an effective algorithm for the continuous m -center problem. The problems defined above are but two examples of a variety of minimax location problems on a network, some of which will be introduced below. The common features of these "center" problems include the network representation and the minimax criterion. The network assumption often allows for more precise modeling in comparison with the use of rectilinear or Euclidean distance metrics. While the minimax criterion is particularly appropriate in emergency service scenarios, other objectives are frequently utilized, for example, the minisum criterion of the "median" problems.

The following categories of model assumptions encompass much of the variety of center problems.

(i) Facility Location Sets. Facilities may be situated on all points of G (vertices and interior points of the arcs) as in Definition 1, or they may be restricted to the set of vertices. In practice, facilities may well be restricted to a given finite set of possible locations.

(ii) Demand Location Sets. In Definition 1 demand is assumed to occur only at the vertices. A natural extension, as we have noted, provides for demand generation at any point on the network.

(iii) Inverse Center Problems. Instead of looking for a set of locations which minimizes the maximum distance for a given number of facilities, it may be appropriate to answer the "inverse" question, namely: What is the least number of facilities and their locations such that the maximum distance from a random incident to the closest facility is less than or equal to a specified value? Referring to such problems as inverse center problems, we can formulate an inverse corresponding to each of the four multicenter problems introduced above. For example, the inverse associated with the absolute m -center problem of Definition 1 is to solve the optimization problem

$$h_\lambda = \min\{h : X_h \subseteq G; l(X_h) \leq \lambda; h \geq 0 \text{ and integer}\}, \quad (4)$$

where h_λ is the minimum number of facilities needed for a maximum distance of λ and $l(\cdot)$ is given by (2).

(iv) **Tree Networks.** An interesting special case of the center problems formulated above occurs when the network contains no cycles, that is, when the network is a tree, say, T . This distinction presents a useful hierarchy in developing computational procedures for the general (cyclic) case. Furthermore, for those physical networks which do exhibit a treelike structure, the special results for trees can be applied directly.

In order to facilitate identification of the center problems formulated above, we shall adopt the shorthand classification scheme shown in Table I. The scheme is best explained by some examples. Consider the absolute m -center problem associated with Definition 1. This would be identified as $A|V|m|G$, since facilities may be established anywhere on the arcs, demand occurs only at the vertices, m centers are to be located, and the network is a general (cyclic) graph. Now consider the inverse center problem defined by (4). That problem is identified as $A|V|\lambda^{-1}|G$, where the " -1 " is used in " λ^{-1} " to distinguish λ from m when numbers replace symbols. As a final example, the 2-center problem on a tree network where facilities and demand can exist anywhere on the network would be denoted $A|A|2|T$.

The absolute m -center problem has been treated extensively ever since Hakimi's seminal paper on the absolute single-center problem [7]. A detailed survey of results in this area can be found in the text by Handler and Mirchandani [10] and in the survey by Tansel et al. [15]. In contradistinction, the continuous m -center problem has hardly been investigated, except for some special cases thereof. Frank [5] was the first to define and solve $A|A|1|G$. Handler solved the tree-based problems $A|A|1|T$ [8] and $A|A|2|T$ [9], and Chandrasekaran and Tamir [3] presented an algorithm for $A|A|m|T$. In [13], Minieka suggested that $A|A|m|G$ was essentially insoluble and instead defined and solved a modified problem in which any link is assumed to be served by one center. In [10], Handler and Mirchandani discuss the applicability of a relaxation algorithm for $A|V|m|G$ to $A|A|m|G$. It is pointed out there that further research is needed to overcome convergence problems. In this paper, we present the results of this research effort.

Section 2 presents the relaxation algorithm for $A|V|m|G$. Proofs and further details may be found in [10]. Section 3 contains a discussion of the direct extension of the relaxation algorithm to $A|A|m|G$, pointing out convergency problems. In Section 4 a modified version of the relaxation algorithm is presented. This algorithm is shown to converge to an optimal solution to $A|A|m|G$, in a finite or infinite number of iterations.

TABLE I. Classification scheme for center problems.

Facility Location Set $\begin{Bmatrix} V \\ A \end{Bmatrix}$	Demand Location Set $\begin{Bmatrix} V \\ A \end{Bmatrix}$	Number of Centers Maximum Distance $\begin{Bmatrix} m \\ \lambda^{-1} \end{Bmatrix}$	Network $\begin{Bmatrix} G \\ T \end{Bmatrix}$
V = vertex set	A = arc set		
m = number of centers	λ = maximum distance		
G = general networks	T = tree network		

A concluding section summarizes the results and indicates the direction for further research efforts.

2. A RELAXATION ALGORITHM FOR THE ABSOLUTE m -CENTER PROBLEM

In this section we describe an efficient methodology for locating m -centers in $A|V|m|G$ and determining the m -radius, r_m . We emphasize here that the algorithm is efficient only in the empirical sense and not as a worst case complexity feature. In fact Kariv and Hakimi [11] showed that the problem is NP-hard. All of the existing algorithms operate on the same principle, generating and solving a series of set-covering problems (CP). The major computational difficulty with these algorithms has to do with the size of the generated CP. The approach we describe here is a problem-oriented relaxation technique which is particularly appropriate for the large-scale CP's encountered in multicenter problems.

We describe first a rudimentary algorithm for $A|V|m|G$ which serves to introduce the CP framework and provides a point of departure for the subsequent developments. Consider first the following definition.

Definition 2. $\langle x, c, y \rangle$, denoting a *local center* at a point $c \in G$ with respect to a pair of vertices $x, y \in V$, is said to exist if the minimum distance from c to x equals the minimum distance from c to y and there is no direction from c in which the minimum distances to both x and y are decreasing. Formally, the conditions are:

$$(i) \quad d(x, c) = d(y, c)$$

$$(ii) \quad \left\{ a \in A_c : \lim_{\theta \rightarrow 0} [(d(z, x_\theta^a) - d(z, c))/\theta] < 0, \text{ for } z = x, y \right\} = \emptyset,$$

where A_c is the set of links or partial links incident at c and x_θ^a is a point distant θ from c on link $a \in A_c$. Note that $\langle x, c, x \rangle$ exists if and only if $c = x$; in this case, the local center $\langle x, x, x \rangle$ is referred to as the *null center* at x .

Let C denote the set of all points in G corresponding to local centers, that is

$$C = \{c \in G : \langle x, c, y \rangle \text{ exists for some } x, y \in V\}. \quad (5)$$

Note that several local centers may coincide at a point $c \in G$, so that the cardinality of C is less than or equal to the number of local centers.

Minieka's Approach

Noting that the cardinality of C is finite, Minieka [12] developed a finite procedure for determining m -center locations. The following result enables the search for m -centers to be restricted from the infinite set of points in G to the finite set in C .

Theorem 1 (Minieka [12]). The finite set C defines a dominant set for the location of centers in $A|V|m|G$, which can now be reformulated as

$$r_m = \min_{X_m \subseteq C} \max_{y \in V} d(X_m, y). \quad (6)$$

A solution strategy for $A|V|m|G$ can now be devised by reformulating (6). Form the matrix $F = \|f_{ij}\|$, with n rows and $|C|$ columns, where f_{ij} denotes the shortest distance from node v_i to the j th candidate center. Then (6) can be formulated as

$$r_m = \min_{H_m \subseteq S} \max_{v_i \in V} \min_{j \in H_m} f_{ij}, \quad (7)$$

where H_m is the index set of a set of m columns in F and $S = \{1, 2, \dots, |C|\}$.

The following algorithm solves (7), and hence $A|V|m|G$, in a finite number of operations.

Algorithm 1 ($A|V|m|G$) (Minieka [12]).

Step 0: Choose an arbitrary initial solution, H_m .

Step 1: (i) Let $d = \max_{v_i \in V} \min_{j \in H_m} f_{ij}$

(ii) Update the matrix $B = \|b_{ij}\|$, where

$$b_{ij} = \begin{cases} 0 & \text{if } f_{ij} \geq d \\ 1 & \text{otherwise.} \end{cases} \quad \begin{matrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, |C| \end{matrix}$$

Step 2: Solve the CP

$$h = \min e'x \quad (8)$$

$$Bx \geq e \quad (9)$$

$$x_i \in \{0,1\}, j = 1, 2, \dots, |C| \quad (10)$$

where $e' = (1, 1, \dots, 1)$.

Step 3: If $h > m$, stop; $r_m = d$ is the value of an optimal solution. Otherwise, an improved solution has been obtained. Update H_m and return to Step 1.

Column Elimination

While the preceding algorithm locates m -centers in a finite number of steps, in practice the procedure is prohibitively time-consuming except for the smallest graphs. As n increases, $|C|$, bounded from above by $n + \binom{n}{2} \cdot |A|$, rapidly becomes extremely large. The result below has the effect of significantly reducing the number of columns in successive iterations of Step 2. Assume for the present that a candidate center $c \in C$ is generated uniquely by just one pair of vertices $x, y \in V$. Relaxing this assumption involves a notational complication which we address later. We define first an important quantity associated with any local center.

Definition 3. For any local center $\langle x, c, y \rangle$, let d_c denote its *range*, defined as

$$d_c = d(x, c) = d(y, c). \quad (11)$$

Theorem 2. In Algorithm 1, if d is the value of the current solution for m -centers, then all candidate centers (columns) in the set $K = \{c \in C : d_c \geq d\}$ can be eliminated without loss of optimality.

The following result renders the CP matrix considerably more sparse. Furthermore, together with Theorem 2, it eliminates the need for storing and working with the matrix F after the algorithm has been initiated.

Theorem 3. In Algorithm 1, for any column j and any row i , the initialization procedure $b_{ij} = 0$ if $f_{ij} > d_j$, where d_j is the range of the candidate center associated with column j , can be made without loss of optimality.

Theorems 2 and 3 lead directly to the following improved version of Algorithm 1.

Algorithm 2 ($A|V|m|G$):

Step 0: (i) Choose an arbitrary initial solution, H_m .

(ii) Let $d = \max_{v_i \in V} \min_{j \in H_m} f_{ij}$.

(iii) Set up matrix $B = \|b_{ij}\|$, where

$$b_{ij} = \begin{cases} 0 & \text{if } f_{ij} > d_j, & i = 1, 2, \dots, n \\ 1 & \text{otherwise.} & j = 1, 2, \dots, |C| \end{cases}$$

Step 1: Eliminate from B all columns j such that $d_j \geq d$.

Step 2: Solve the CP ((8)–(10)).

Step 3: If $h > m$, stop; $r_m = d$ is the value of an optimal solution. Otherwise an improved solution has been obtained. Update H_m ; let $d = \max_{j \in H_m} d_j$ and return to Step 1.

In the revised algorithm columns of B are eliminated, whereas previously elements of B were zeroed, leaving B , in general, with its original dimensions. The effect of this reduction in size becomes increasingly significant as d diminishes with increasing m . Notice, however, that for small m we still encounter a great number of columns which, coupled with large n , renders the algorithm impractical for large problems. Before we proceed to tackle this issue, we need to relax the restrictive assumption made earlier that only one local center may exist at a point in G .

Consider now the general case where k_c local centers $\{\langle x_i, c, y_i \rangle\}_{i=1}^{k_c}$ coincide at a point $c \in G$. Similar to C of (5), let C' denote the set of all local centers, that is

$$C' = \{\langle x_i, c, y_i \rangle : i = 1, 2, \dots, k_c, \quad \text{for all } c \in C\}. \quad (12)$$

To generalize the column elimination procedure to this case, assume that the set C in Theorems 1 and 2 is replaced by C' , so that a single point in G may be associated with several candidate centers.

The theory and algorithm apply as before. Notice, however, that the number of columns in the CP matrices may exceed the number of columns in Miniéka's procedure. In particular, the initial B matrix now contains $|C'| \geq |C|$ columns. However, it is simple to show that all local centers at a point on G may be represented by at most one column in the CP matrix, and hence the CP matrix need never contain more than $|C|$ columns.

Suppose that after Step 1 of the algorithm the matrix B contains a set of k columns corresponding to k local centers at a point in G . Suppose the columns are indexed $j = 1, 2, \dots, k$ so that $d_1 \geq d_j$, $j = 2, 3, \dots, k$. Then, by construction, $b_{i1} \geq b_{ij}$, $i = 1, 2, \dots, n$, $j = 2, 3, \dots, k$, and the structure of (9) renders columns $j = 2, 3, \dots, k$ redundant in the CP. Notice, however, that these columns must be stored for future use in case column 1 is eliminated at some later stage.

A Relaxation Strategy

The size of the CP matrices arising in Algorithm 1 is n rows by $|C|$ columns. For networks with hundreds of nodes, the corresponding matrices are very large indeed. While in Algorithm 2 the number of columns is generally reduced, the CP matrices may still be large, particularly for small values of m . In both algorithms, solving the CP's in Step 2 is a major computational burden severely limiting the size of problems which can be optimally solved.

Comprehensive reviews of CP's appear in Garfinkel and Nemhauser [6] and Christofides and Korman [4]. While significant algorithmic advances have been achieved (e.g., Balas [1], Balas and Ho [2]), the state of the art does not permit efficient solution of truly large-scale general CP's. The following strategy for multicenter location constitutes a problem-oriented technique for solving the very large CP's arising in this context.

The "relaxation strategy" for $A|V|m|G$ is based upon two fundamental observations:

(i) For any number of centers, m , there exists a set of critical vertices, a "relaxed" set, $R \subseteq V$, which determine the optimal location of centers. Moreover, $|R|$ will be relatively unaffected by n , and instead will be fairly closely related to m .

In relation to other mathematical programming problems, this observation and implied strategy is akin to a relaxation approach for minimax formulations where, again, the observation is that most of the constraints in the derived constrained problem will be nonbinding at an optimal solution.

(ii) Unlike general mathematical programming relaxation strategies, in this case both rows and columns are eliminated as a result of the manner in which columns (candidate centers) are generated by rows (vertices). Thus, a special feature here is that advantages which generally result from restriction are obtained as well, rendering the relaxation strategy particularly effective.

The following list of definitions together with previously defined quantities identifies notation used in the relaxation algorithm. We shall assume that a candidate center $c \in G$ is generated uniquely by one pair of vertices. As before, this restriction may be relaxed by replacing C of (5) with C' of (12).

$$C(x,y) = \{c \in C: \langle x,c,y \rangle \text{ exists for a given pair } x,y \in V\}.$$

$$C[(x,y):d] = \{c \in C(x,y): d_c < d\}.$$

$$C\{R:d\} = \bigcup_{x,y \in R} C[(x,y):d].$$

$$\{h, \hat{X}_h, \hat{d}\} = \text{optimal solution quantities for the CP, where}$$

$$h = \text{value of objective function}$$

$$\hat{X}_h = \text{identity of the } h \text{ centers in the solution}$$

$$\hat{d} = \max_{c \in \hat{X}_h} \{d_c\}.$$

Following is a statement of a procedure for determining m -centers, which combines

Algorithms 1 and 2 in a relaxation framework. We shall state the algorithm in the context of solving $A|V|m|G$ for $m = 1, 2, \dots, M$.

Algorithm 3 ($A|V|m|G$):

Step 0 Initialization: Set $m = 1$ and select arbitrarily a vertex x . Let $R = \{x\}$, $d = \infty$, $C\{R:d\} = \langle x, x, x \rangle$, $\hat{d} = 0$, $\hat{X}_1 = x$. Proceed to Step 1.

Step 1 Improvement/Vertex-to-Enter: If every vertex in $V-R$ is within a range of $\hat{d}$ from some $x \in \hat{X}_m$, then $\hat{X}_m$ is an improved solution; let $X_m = \hat{X}_m$ be the new current solution with $d = \hat{d}$ an upper bound on r_m ; update $C\{R:d\}$ by eliminating candidate centers whose ranges exceed or equal d and go to Step 3. Otherwise, designate as vertex-to-enter a vertex v , which is furthest away from $\hat{X}_m$. If $\hat{d} < l(\hat{X}_m) < d$, then $\hat{X}_m$ is still an improved solution; set $d = l(\hat{X}_m)$ and let $X_m = \hat{X}_m$. In both cases proceed to Step 2.

Step 2 Column Generation: Add v to R and update $C\{R:d\}$ by adding candidate centers $C[(v,r):d]$ for all $r \in R$. Proceed to Step 3.

Step 3 Covering Problem: Select from $C\{R:d\}$ an arbitrary set of m candidate centers, $\hat{X}_m$, "covering" all vertices in R and go to Step 1. If this is impossible, proceed to Step 4.

Step 4 Optimality: The current solution X_m , constitutes a set of m -centers, with radius $r_m = d$. If $m = M$, stop. Otherwise, let $m = m + 1$ and return to Step 3.

Algorithmic details, particularly those pertaining to column generation, are given in [10]. The reader is referred to that reference also for examples and for computational results. Summarizing those results briefly, it appears that the relaxation algorithm is well-suited to solving $A|V|m|G$ for large-scale networks.

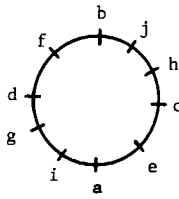
3. DIRECT EXTENSION OF RELAXATION ALGORITHM TO $A|A|m|G$

The continuous m -center problem is perhaps the most challenging of the variety of minimax network location problems. Physically, it represents what is often the most realistic situation while $A|V|m|G$ is some discretized approximation of it. One way of approaching this problem is indeed to discretize the set of demand generating points as finely as is desired and then solve as $A|V|m|G$. Without the relaxation technique, such a scheme becomes unwieldy as the approximation is tightened because of the critical effect of a large number of vertices in other techniques.

Consider now the relaxation approach described in the previous section. All along we have been solving relaxed problems, $A|R|m|G$, $R \subseteq V$, as surrogates for $A|V|m|G$. The only conceptual change here is that V is now the infinite set of points in G . The basic strategy remains unaltered.

In Algorithm 3, $l(X_m)$ is now changed to $l'(X_m)$ as defined in (3) and vertex-to-enter is changed to point-to-enter (pte).

Although $A|A|m|G$ utilizes to the fullest extent the capabilities of the relaxation method and provides the most striking context for its application, the algorithm nevertheless remains elusive in its convergence properties when applied to $A|A|m|G$. We know that for $A|V|m|G$ the algorithm converges finitely to an optimal solution [10].


 FIG. 1. A single cycle network to illustrate $A|A|1|G$.

If the algorithm converges finitely for $A|A|m|G$, then the solution is optimal. However, the algorithm may not be finite for $A|A|m|G$. The following example illustrates this. Consider the network in Figure 1 composed of a single cycle, and solve the problem $A|A|1|G$. The application of the algorithm beginning at some arbitrary point, a , will lead to an infinitely convergent series of intermediate solutions. The labels $a, b, c, \dots$ in Figure 1 represent successive additions to R as determined by (3). Successive iterations of the algorithm yield intermediate results as shown in Table II.

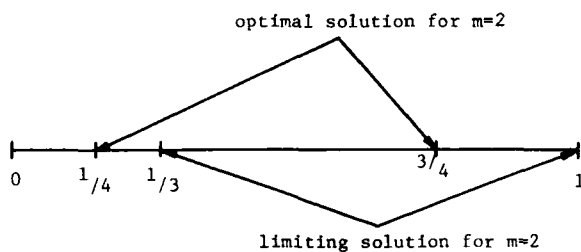
In this example, of course, any point on the cycle is a bona fide single-center. However, the algorithm has difficulty in recognizing optimality. It will only do so in a limiting sense.

When the algorithm fails to terminate finitely, it is possible that the limiting solution will not be optimal. To illustrate this, consider the problem of locating continuous 2-centers on a graph composed of a single link of length 1, as in Figure 2. Denote points on the line by their distance from the left endpoint. Solving first for $m = 1$, we find the optimal solution at $\frac{1}{2}$. The quantities at the end of this stage are: $R = \{0, 1\}$, $d = \frac{1}{2}$, $C\{R:d\} = \{0, 1\}$, $X_1 = \frac{1}{2}$. Solving now for $m = 2$, we initiate the search with the quantities $\hat{X}_2 = \{0, 1\}$, $\hat{d} = 0$, $l'(\hat{X}_2) = \frac{1}{2}$, $\text{pte} = \frac{1}{2}$, $R = \{0, \frac{1}{2}, 1\}$, $C\{R:\frac{1}{2}\} = \{0, \frac{1}{4}, \frac{3}{4}, 1\}$, which yields the CP matrix:

$$\begin{array}{c} 0 \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \\ d_c [0 \quad \frac{1}{4} \quad 0 \quad \frac{1}{4} \quad 0]. \end{array}$$

 TABLE II. Intermediate results for $A|A|1|G$ in the single cycle network example.

R	$\hat{X}_1$	$2\hat{d}$	$2l'(\hat{X}_1)$
a	a	0	1
a, b	d	$\frac{1}{2}$	1
a, b, c	c	$\frac{1}{2}$	1
a, b, c, d	f	$\frac{3}{4}$	1
a, b, c, d, e	e	$\frac{3}{4}$	1
a, b, c, d, e, f	h	$\frac{3}{4}$	1
a, b, c, d, e, f, g	g	$\frac{3}{4}$	1
a, b, c, d, e, f, g, h	j	$\frac{7}{8}$	1
etc.			

FIG. 2. A single link network to illustrate $A|A|2|G$.

The CP contains three possible solutions, of which we choose $\hat{X}_2 = \{\frac{1}{4}, 1\}$, $\hat{d} = \frac{1}{4}$, $l'(\hat{X}_2) = \frac{3}{8}$, $\text{pte} = \frac{5}{8}$, $R = \{0, \frac{1}{2}, \frac{5}{8}, 1\}$, $C\{R: \frac{1}{2}\} = \{0, \frac{1}{4}, \frac{5}{16}, \frac{1}{2}, \frac{9}{16}, \frac{5}{8}, \frac{3}{4}, \frac{13}{8}, 1\}$ which yields the CP matrix:

$$0 \begin{bmatrix} 1 & 1 & 1 \\ \frac{1}{2} & & 1 & 1 & 1 & 1 \\ \frac{5}{8} & & 1 & & 1 & 1 & 1 \\ 1 & & & & 1 & 1 & 1 \end{bmatrix}$$

$$d_c [0 \quad \frac{1}{4} \quad \frac{5}{16} \quad 0 \quad \frac{1}{16} \quad 0 \quad \frac{1}{4} \quad \frac{3}{16} \quad 0].$$

Choosing $\hat{X}_2 = \{\frac{5}{16}, 1\}$ we find $\hat{d} = \frac{5}{16}$, $l'(\hat{X}_2) = \frac{11}{32}$, $\text{pte} = \frac{31}{32}$. Continuing in this manner, we can obtain an infinite series of solutions as indicated in Table III, where

$$a_n = \sum_{i=0}^{n-1} 4^i / 4^n \quad n \geq 1$$

$$= \frac{1}{3} - \frac{1}{3 \cdot 4^n}.$$

We thus find $\lim_{n \rightarrow \infty} a_n = \frac{1}{3}$. The limiting solution $\hat{X}_2 = \{\frac{1}{3}, 1\}$ yields $\hat{d} = \frac{1}{3}$, $l'(X_2) = \frac{1}{3}$, $\text{pte} = \frac{2}{3}$, while the optimal solution is $X_2^* = (\frac{1}{4}, \frac{3}{4})$, $l'(X_2^*) = \frac{1}{4}$.

We have seen an example where the relaxation algorithm applied to $A|A|m|G$ converges infinitely to a suboptimal solution. In conclusion, the relaxation algorithm solves

TABLE III. Intermediate results for $A|A|2|G$ in the single link network example.

n	$\hat{X}_2$	$\hat{d}$	pte
0	$\{0, 1\}$	0	$\frac{1}{2}$
1	$\{\frac{1}{4}, 1\}$	$\frac{1}{4}$	$\frac{5}{8}$
2	$\{\frac{5}{16}, 1\}$	$\frac{5}{16}$	$\frac{11}{32}$
3	$\{\frac{21}{64}, 1\}$	$\frac{21}{64}$	$\frac{25}{128}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$\{a_n, 1\}$	a_n	$2a_{n+1}$

$A|A|m|G$ if it converges finitely. Otherwise, we need to develop a different strategy to solve $A|A|m|G$.

4. MODIFIED RELAXATION ALGORITHM FOR $A|A|m|G$

In this section we present a modified version of Algorithm 3 which guarantees convergence to an optimal solution for $A|A|m|G$. The modification is to optimally solve the relaxed problem $A|R|m|G$ for every R . In terms of Algorithm 3 this implies a modification of Step 3 to read:

Step 3 Covering Problem: Select from $C\{R:d\}$ a set of m candidate centers $\hat{X}_m$ "covering" all points in R with minimal $\hat{d}$ and go to Step 1. If this is impossible, proceed to Step 4.

Implementation of the new Step 3 involves in general a finite series of old Step 3's. We begin by identifying, if possible, an arbitrary set $\hat{X}_m$ covering the points in R . We then delete from $C\{R:d\}$ all columns with $d_c \geq \hat{d}$. We continue in this manner till it is impossible to cover R , so that the last set $\hat{X}_m$ represents an optimal solution to $A|R|m|G$. With this solution, we proceed to Step 1. It is important to observe, however, that the algorithm continues with the original set of candidate centers $C\{R:d\}$. Finally, we note that Algorithm 3 is also modified to apply to $A|A|m|G$, as specified in Section 3.

The modified algorithm may be viewed as a generalization of the original algorithm since it can solve not only $A|A|m|G$ but $A|V|m|G$ as well. The latter fact is obvious since in the modified algorithm, Step 3 is merely refined to reduce the degree of freedom in the original Step 3. We now proceed to prove the assertion that the modified algorithm solves $A|A|m|G$.

Let r_m be the m -radius in $A|A|m|G$ and let r_m^n be the m -radius in $A|R|m|G$ where the cardinality of R is n . By r_m^{n+1} , we refer to the m -radius in $A|R'|m|G$ where $R' = R \cup \{y\}$, $y \in G$. We state without proof the fact that the modified relaxation algorithm yields r_m^n at every iteration of Step 3. The proof of this reverts to the proof of Algorithm 3 as given in [10]. To prove that $\lim_{n \rightarrow \infty} r_m^n = r_m$ we need several lemmas.

Lemma 1. $r_m^n \leq r_m^{n+1} \quad \forall n$.

This lemma is obviously true and does not require formal proof.

Lemma 2. The sequence of bounds $\{d^K\}_1^\infty$ in successive iterations of the modified algorithm is a nonincreasing sequence.

Proof. From the algorithm

$$d^{K+1} = \begin{cases} l^{K'}(\hat{X}_m) & \text{if } l^{K'}(\hat{X}_m) < d^K \\ d^K & \text{otherwise} \end{cases} \quad K = 0, 1, 2, \dots$$

$$d^0 = \infty. \quad \blacksquare$$

Lemma 3. The sequence of bounds $\{d^K\}_1^\infty$ is bounded from above $d^K \leq d^1 < \infty$; $K = 2, 3, \dots$

Proof. Initially $d^0 = \infty$ and $\hat{d} = 0$, so that a pte is introduced to R and $d^1 = l^{1'}(\hat{X}_m) < \infty$. ■

Lemma 4. The sequence $\{r_m^n\}_{n=0}^\infty$ is bounded from above.

Proof. From the algorithm $r_m^n \leq d^n \forall n$. And from Lemmas 2 and 3, we have $r_m^n \leq d^1 < \infty$. ■

Lemma 5. $\{r_m^n\}_{n=0}^\infty$ is a convergent sequence. The lemma follows from Lemmas 1 and 4.

Lemma 6. At every iteration of the modified algorithm

$$r_m^n \leq r_m \leq l^{n'}(\hat{X}_m).$$

Both sides of the expression $r_m^n \leq r_m$ and $r_m \leq l^{n'}(\hat{X}_m)$ obviously hold.

Lemma 7. $\lim_{n \rightarrow \infty} (l^{n'}(\hat{X}_m) - r_m^n) = 0$.

Proof. Suppose on the contrary that

$$\lim_{n \rightarrow \infty} (l^{n'}(\hat{X}_m) - r_m^n) = \delta > 0,$$

and consider the situation arising at each iteration of the algorithm. In Step 3, a new pte is added to R . Two cases are discernible: (i) pte $\in V$; (ii) pte $\notin V$. Clearly, case (i) can occur only a finite number of times so that for n sufficiently large all pte's are of case (ii). We shall now demonstrate that case (ii) also can only occur a finite number of times. Consider a generic case (ii) as illustrated in Figure 3. The shaded portions represent segments covered by $\hat{X}_m$ within radius r_m^n . Note that for sufficiently large n (i.e., $n > m$), $r_m^n > 0$. The arrows point away from the center. The point farthest away, pte, is located at the midpoint of an interval at least 2δ in length which represents points outside the radius r_m^n . Since pte is now incorporated into R , it will be covered in all future iterations. The center covering pte will lie (a) outside the 2δ interval, or (b) inside it. We wish to determine the amount of the 2δ interval covered, together with pte, in all future iterations. Denote this quantity I .

Case a (center covering pte outside the interval): $I \geq \delta$.

Case b (center covering pte inside interval): $I \geq \min\{\delta, r_m^n\}$.

According to our original supposition, every iteration of the algorithm will yield (in case (ii)) at least one uncovered interval of size greater than or equal to 2δ . And for each of these intervals a portion $I \geq \min\{\delta, r_m^n\}$ will be covered in future iterations. Summing these, we obtain an infinite network. This contradiction implies that case

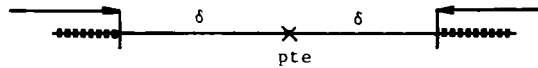


FIG. 3. Case (ii) for pte.

(ii) can only occur a finite number of times so that the original supposition cannot hold and the lemma is established. ■

We are now ready to state and prove Theorem 4.

Theorem 4. The modified relaxation algorithm converges to an optimal solution in $A|A|m|G$.

Proof. When convergence is finite, proof of optimality is essentially the same as for $A|V|m|G$ as given in [10]. For infinite convergence, we find from Lemmas 5, 6, and 7

$$\lim_{n \rightarrow \infty} l^{n'}(\hat{X}_m) = \lim_{n \rightarrow \infty} r_m^n = r_m. \quad \blacksquare$$

Note. Employing the original Algorithm 3 (as extended to $A|A|m|G$ in Section 3), we would derive the following versions of Lemmas 6 and 7,

$$\text{L.6: } \hat{r}_m^n \geq r_m^n \leq r_m \leq l^{n'}(\hat{X}_m)$$

where $\hat{r}_m^n$ is the current upper bound on r_m^n in Step 3, and

$$\text{L.7: } \lim_{n \rightarrow \infty} l^{n'}(\hat{X}_m) = \lim_{n \rightarrow \infty} \hat{r}_m^n$$

so that no assertion can be made regarding r_m from limiting quantities of the algorithm.

5. CONCLUSION

A relaxation algorithm has been described for the continuous m -center problem. The algorithm can be conceptualized as a column (candidate centers) and row (points to be covered) generating procedure for the infinite-size covering problems needed to solve the problem. The discussion has been primarily theoretical. Details of the covering problems and minimum path subroutines have not been included. In particular, the computation of $l'(\hat{X}_m)$ is given in [14]. The most important sequel to this effort is to gain computational experience with the algorithm. Based on similar experience with the absolute m -center problem, we feel fairly confident that the procedure will be practical for large-scale problems. Another sequel to this paper would be an attempt to refine the algorithm so as to assure finite convergence.

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